SPECTRAL EXPRESSION OF A PERIODIC ORBIT USING CHEBYSHEV POLYNOMIALS

with the examples of the Lorenz model and a differentially heated cavity flow.

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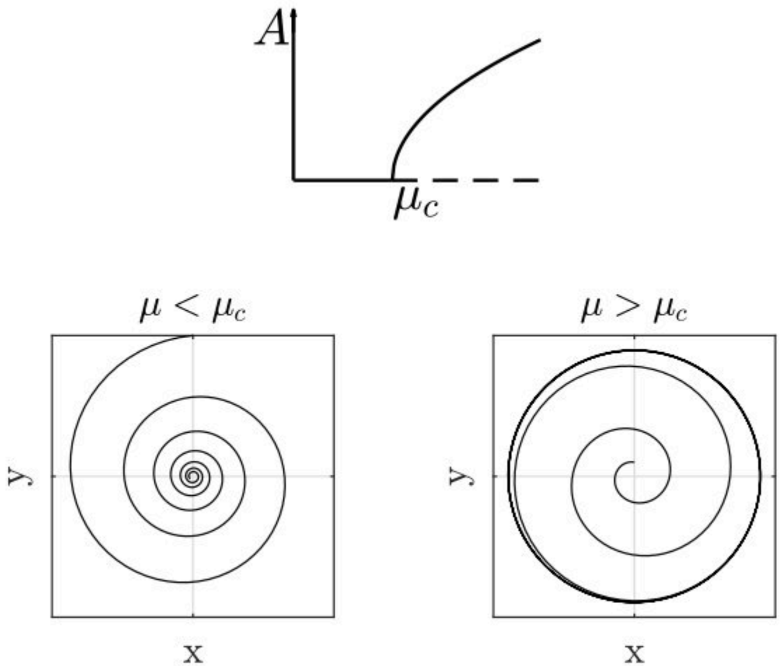
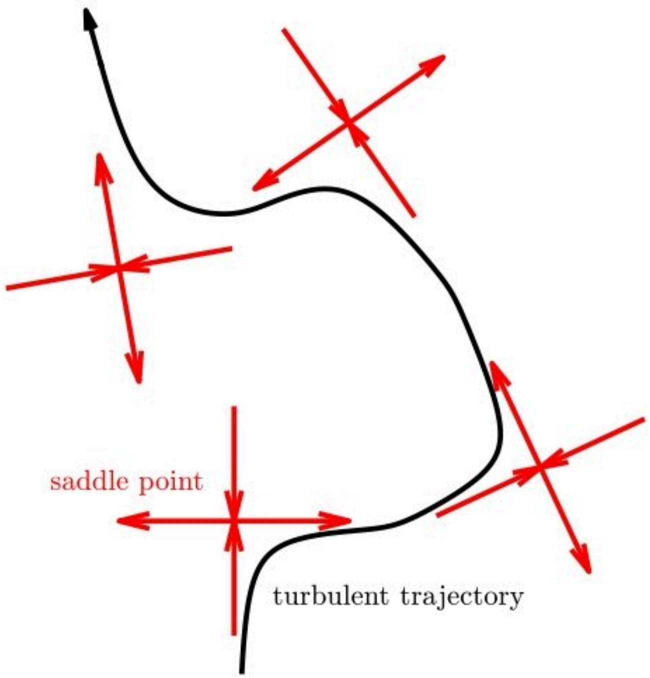
Why do we look for periodic orbits?

They are the simplest form of unsteadiness in a system.



Supercritical Hopf bifurcation. A stable limit cycle emanates from the critical point μ.

They are believed to be the backbone of the chaotic dynamics.



Chaotic trajectory travels between saddle points in the phase space.

Problem statement

Given a **nonlinear autonomous** system:

$$\dot{\boldsymbol{x}} = \boldsymbol{f}(\boldsymbol{x})$$

for example:

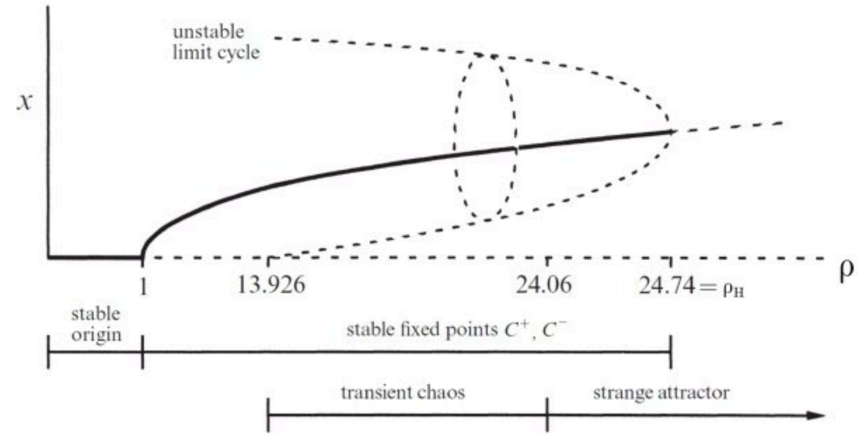
$$\dot{x} = \sigma(y - x)$$

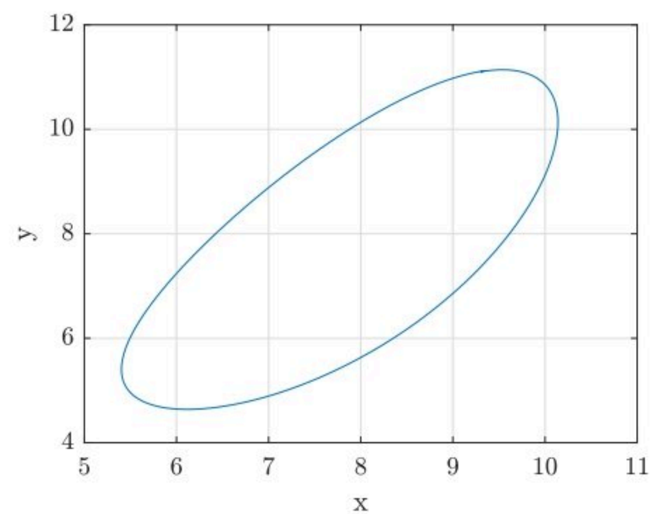
$$\dot{y} = x(\rho - z) - y$$

$$\dot{z} = xy - \beta z$$

What is a periodic orbit of the dynamical system?

$$\boldsymbol{x}(t + T) = \boldsymbol{x}(t)$$

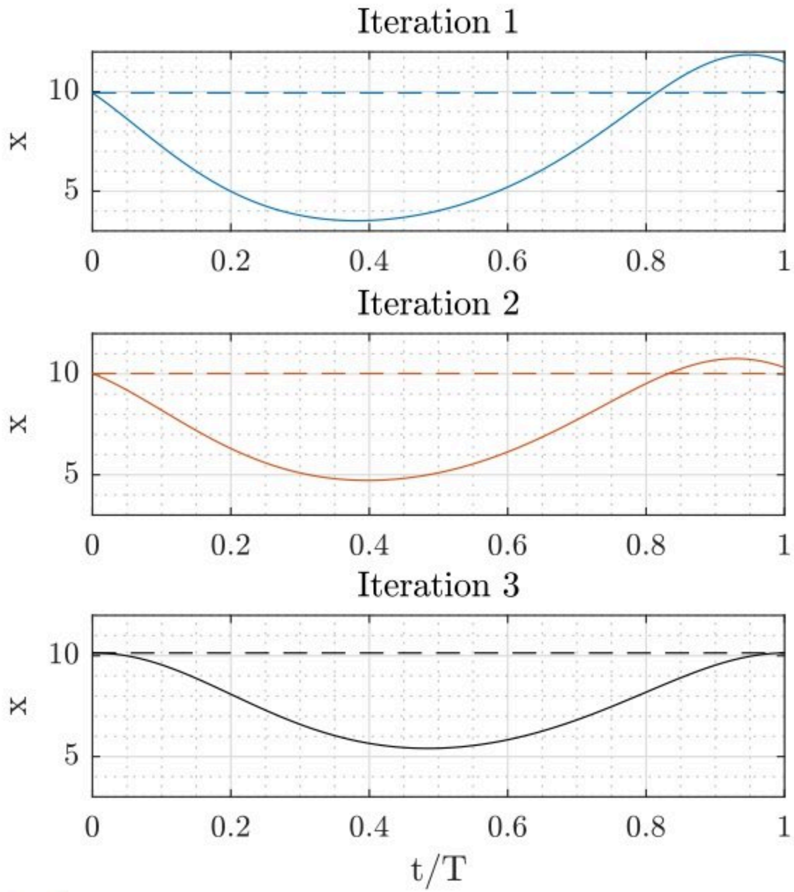




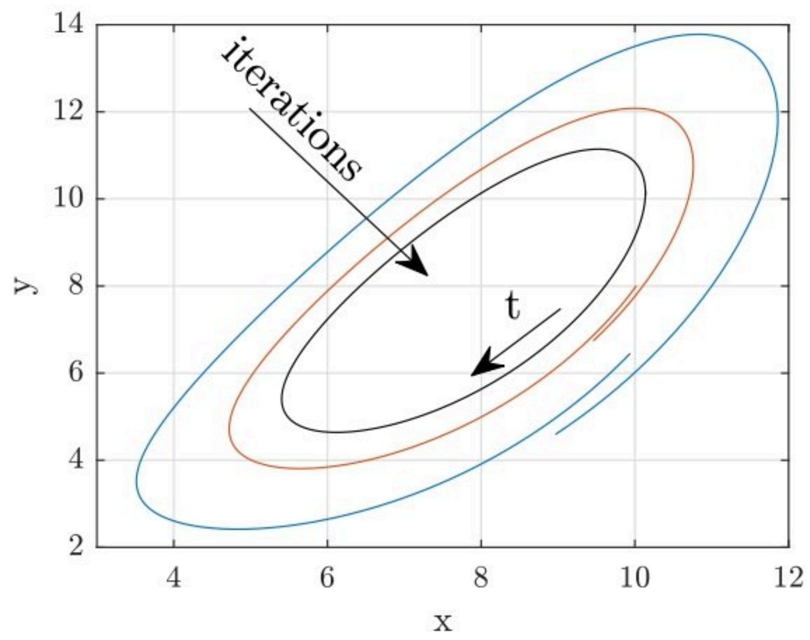
Lorenz system, $\rho = 24$, $\sigma = 10$, $\beta = 8/3$, $\boldsymbol{x}_0 = (9.3, 11.1, 22.2)$ [Strogatz]

Shooting method

Time integration:

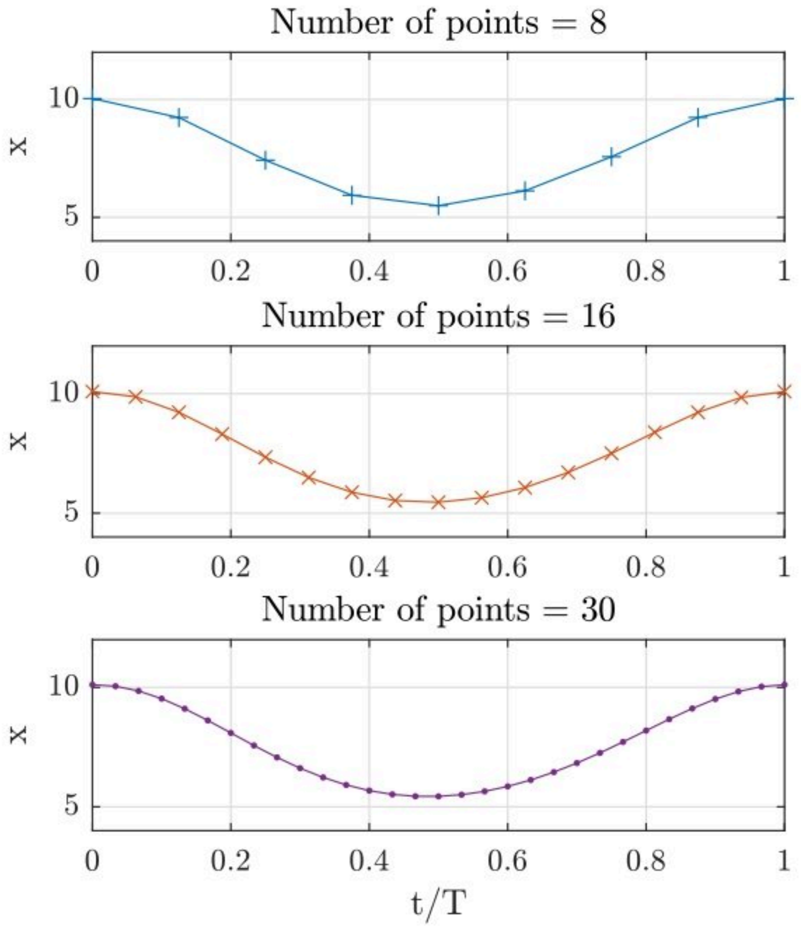


Method:
refine $\boldsymbol{x}_0$ until $\boldsymbol{x}(t+T) = \boldsymbol{x}(t)$.

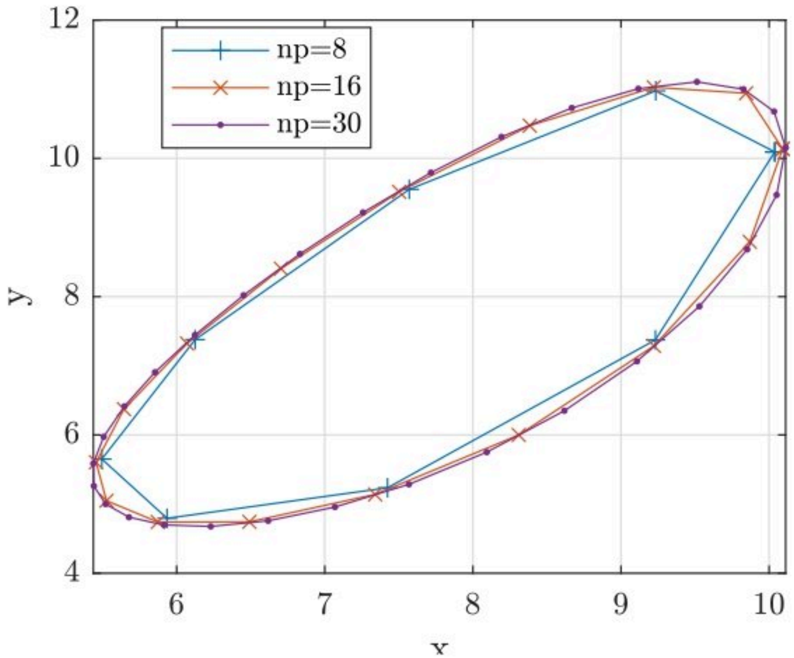


Unknowns:
$\boldsymbol{x}_0 = [x_0,\ y_0,\ z_0]$ and T

Lorenz system, $\rho = 24$, $\sigma = 10$, $\beta = 8/3$



Collocation method:



Lorenz system, $\rho = 24$, $\sigma = 10$, $\beta = 8/3$

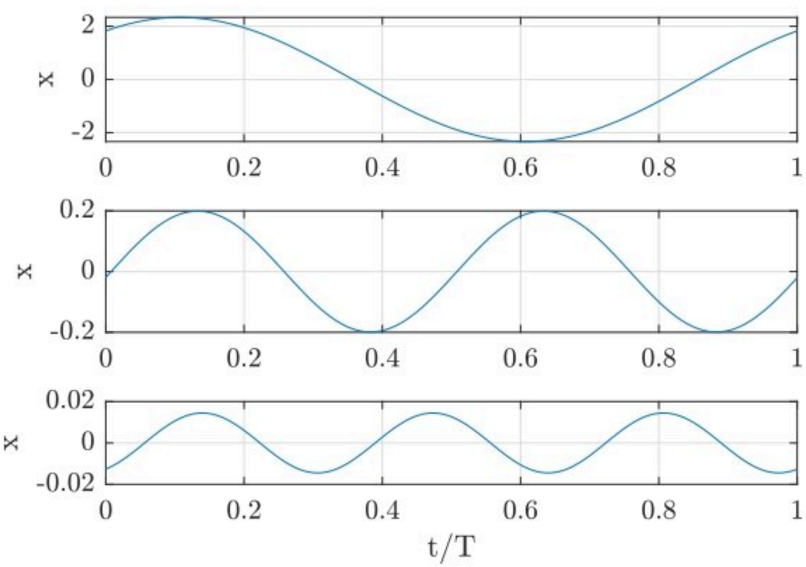
Vector of unknowns: $\boldsymbol{u} = [x_0,\ y_0,\ z_0,\ x_1,\ y_1,\ z_1,\ \dots,\ T]$

We look for such an ensemble of snapshots $\boldsymbol{x}_i = [x_i, y_i, z_i]$ that would satisfy the governing equations.

Spectral formulation

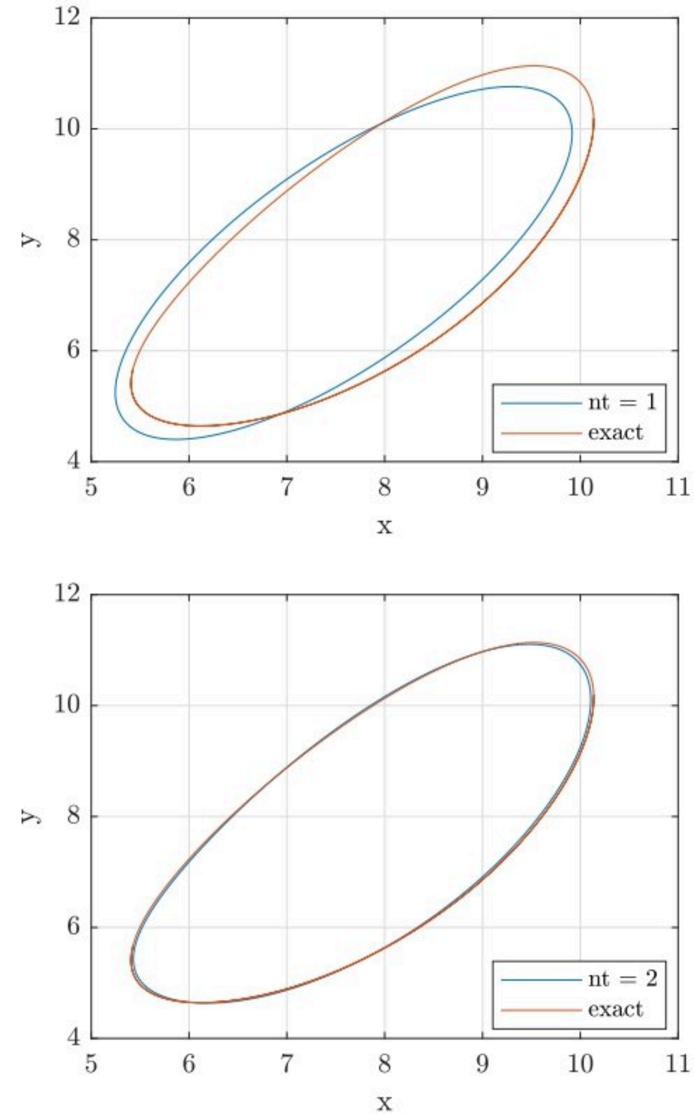
Variables are spectral coefficients:

$$x(t) = \sum_{k=-\infty}^{\infty} u_k \phi_k(t) \approx \sum_{k=-nt}^{nt} u_k e^{ik\omega t}$$



Lorenz system, $\rho = 24$, first 3 harmonics

Variables: $[u_{-nt}, u_{-nt+1}, \ldots, u_{nt}, \omega]$



Lorenz system, $\rho = 24$

How to find the spectral expression of a limit cycle?

Insert $x(t) = \sum_{k=-nt}^{nt} u_k \phi_k(t)$ into the governing equations:

$$\dot{x} = \sigma(y - x)$$

$$\dot{y} = x(\rho - z) - y$$

$$\dot{z} = xy - \beta z$$

use Galerkin method and a phase condition to generate the set of equations $\boldsymbol{g}_{[3(2nt+1)+1]\times 1}(\boldsymbol{u})$. Additional variable is the period of the orbit.

Phase condition can be:

$$\dot{x}(t = 0) = 0$$

Linearise $\boldsymbol{g}$ around $\boldsymbol{u}$ and apply the Newton method to find zeros of $\boldsymbol{g}$:

$$\frac{\partial \boldsymbol{g}}{\partial \boldsymbol{u}} \boldsymbol{\delta u} = \boldsymbol{J} \boldsymbol{\delta u} = -\boldsymbol{g}(\boldsymbol{u})$$

$$\boldsymbol{u} = \boldsymbol{u} + \boldsymbol{\delta u}$$

Upon few iterations we find the vector of spectral coefficients $\boldsymbol{u}$ and the period of the orbit.

Time dependence expressed with Fourier modes - Harmonic balance

Let's take a simple equation:

$$\dot{x} = x^2, \qquad x = \sum_{k=-1}^{1} u_k e^{ik\omega t}$$

$$-i\omega u_{-1} e^{-i\omega t} + 0 \cdot i\omega u_0 + i\omega u_1 e^{i\omega t} =$$

$$u_{-1}u_{-1}e^{-2i\omega t} +$$

$$(u_{-1}u_0 + u_0 u_{-1})e^{-i\omega t} +$$

$$u_{-1}u_1 + u_0 u_0 + u_1 u_{-1} +$$

$$(u_1 u_0 + u_0 u_1)e^{i\omega t} +$$

$$u_1 u_1 e^{2i\omega t}$$

We can use Galerkin method and orthogonality of Fourier modes to generate:

$$0 = i\omega u_{-1} + u_{-1}u_0 + u_0 u_{-1}$$

$$0 = u_{-1}u_1 + u_0 u_0 + u_1 u_{-1}$$

$$0 = -i\omega u_1 + u_1 u_0 + u_0 u_1$$

+ phase condition:

$$0 = -i\omega u_{-1} + i\omega u_1$$

this gives a vector $\boldsymbol{g}_{4\times 1}(\boldsymbol{u})$. We find u_k and ω using Newton method.

Stability of the solution - Hill's method

We introduce a small perturbation to our solution:

$$x + x_{pert} = \sum_{k=-1}^{1} (u_k + v_k e^{\mu t}) e^{ik\omega t}$$

to get:

$$\mu v_{-1} = i\omega v_{-1} + v_{-1} u_0 + v_0 u_{-1}$$

$$\mu v_0 = v_{-1} u_1 + v_0 u_0 + v_1 u_{-1}$$

$$\mu v_1 = -i\omega v_1 + v_1 u_0 + v_0 u_1$$

This yields an eigenvalue problem:

$$\mu \boldsymbol{v} = \boldsymbol{J} \boldsymbol{v}$$

Eigenvalue $\mu \in \mathbb{C}$ is a Floquet exponent of the periodic orbit. Floquet multiplier is defined as:

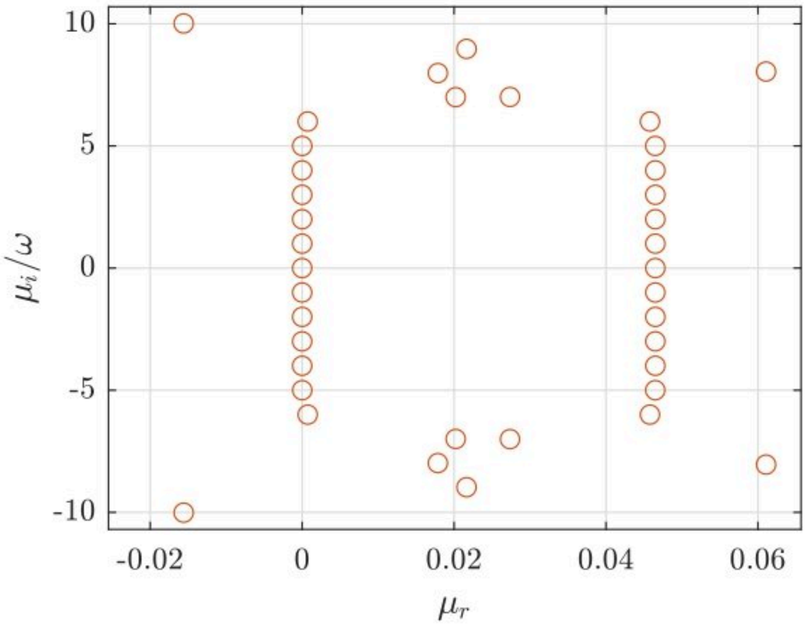
$$\frac{x_{pert}(t = T)}{x_{pert}(t = 0)} = e^{\mu T} = \sigma$$

and conversely:

$$\mu = \frac{\log \sigma}{T}$$

Instability if:
$real(\mu) > 0$ ($|\sigma| > 1$)

$$\mu \boldsymbol{v} = \boldsymbol{J}\boldsymbol{v}$$



Zoom on the spectrum of Hill's matrix ($nt = 20$) for the unstable orbit of Lorenz system $\rho = 24$.

Interpretation problem:
Many more eigenvalues than dof=3.
$nt \to \infty$ $\mu \pm in\omega$ ($n \in \mathbb{N}$) is an eigenvalue.

$$x + x_{pert} = \sum_{k=-nt}^{nt} (u_k + v_k e^{\mu t}) e^{ik\omega t}$$

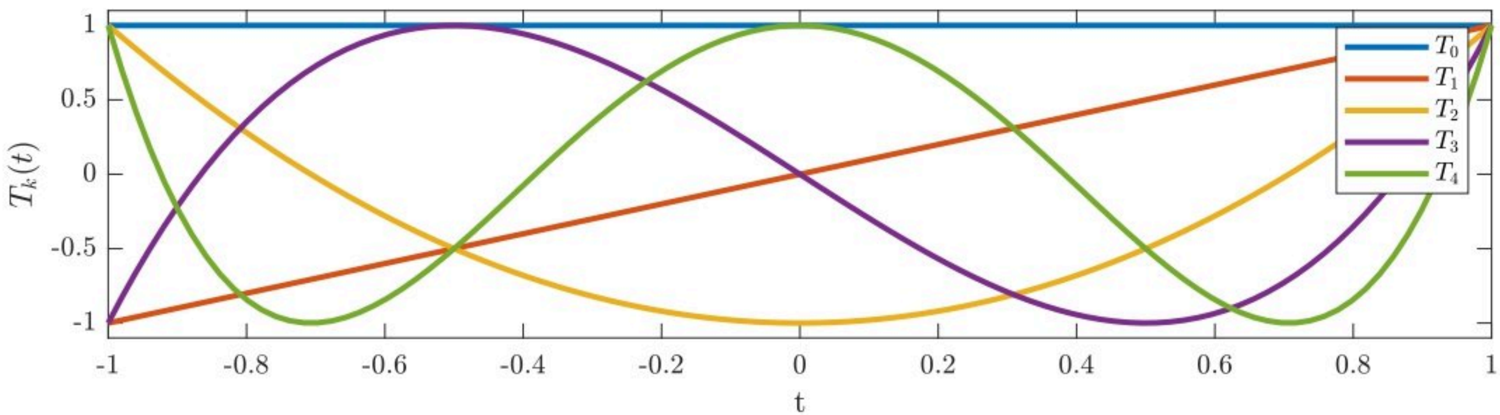
Problem:
- periodic basis to describe a temporally growing perturbation
- finite nt

Chebyshev dependence in time

Motivation: Hill's method yields μ which is not unique.

When $nt \to \infty$ $\mu \pm in\omega$ ($n \in \mathbb{N}$) is also an eigenvalue.

$$x = \sum_{k=0}^{cnt} u_k T_k(t) = \sum_{k=0}^{cnt} u_k \cos(k \arccos t), \quad t \in (-1, 1)$$



Chebyshev polynomials do not enforce periodicity:

$x(1) = x(-1)$ for a periodic orbit and

$x_{pert}(1) = \sigma x_{pert}(-1)$ will have to be enforced.

Stability of the solution - Chebyshev expansion

We introduce a small perturbation to the solution:

$$x + x_{pert} = \sum_{k=0}^{1} (u_k + v_k) T_k(t)$$

to $0 = \dot{x} - x^2$ to get:

$$0 = v_0 \dot{T}_0(t) + v_1 \dot{T}_1(t) +$$
$$-2 v_0 u_0 T_0(t) T_0(t) - 2 v_0 u_1 T_0(t) T_1(t)$$
$$-2 v_1 u_0 T_1(t) T_0(t) - 2 v_1 u_1 T_1(t) T_1(t)$$

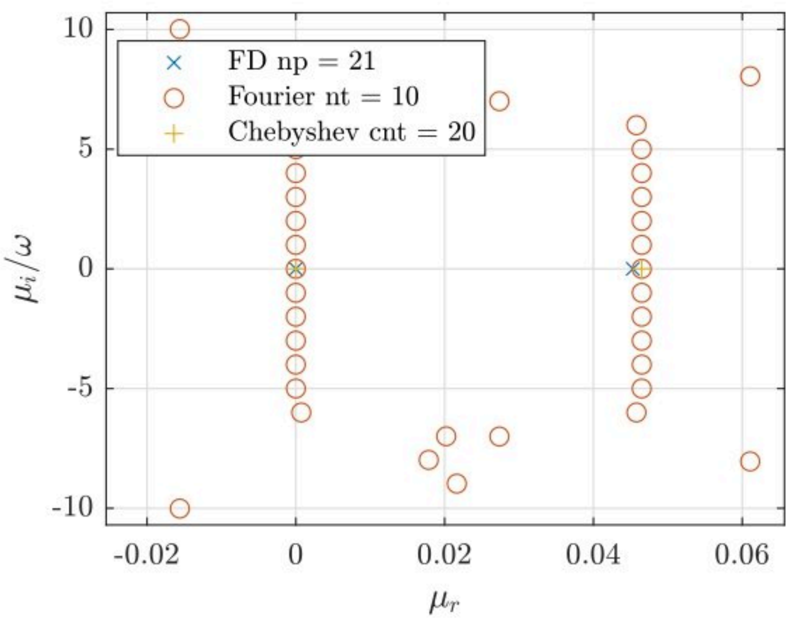
$$\sum_{k=0}^{1} v_k T_k(1) = \sigma \sum_{k=0}^{1} v_k T_k(-1)$$

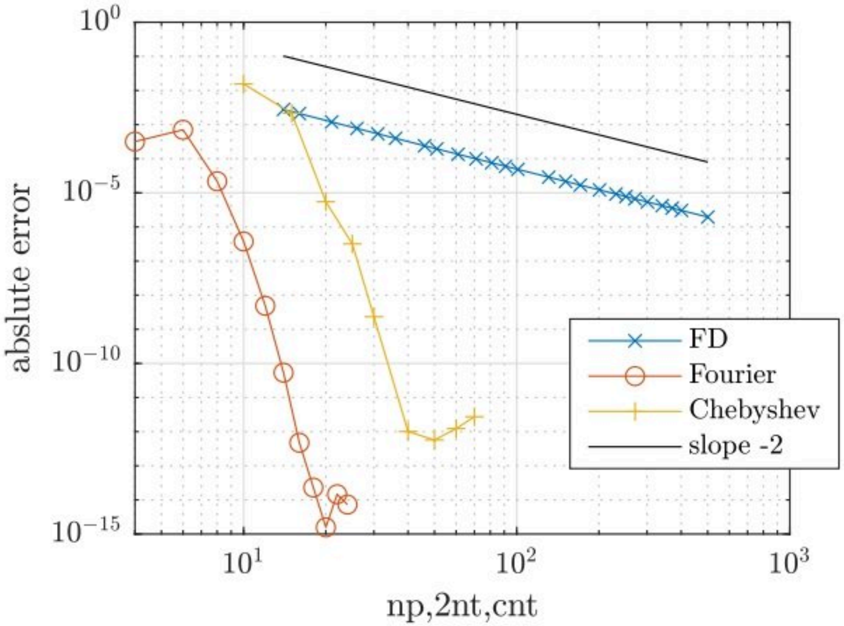
Evaluation at the Gauss-Lobatto collocation points yields:

$$\sigma \boldsymbol{B} \boldsymbol{v} = \boldsymbol{J} \boldsymbol{v}$$

Eigenvalue $\sigma \in \mathbb{C}$ is a Floquet **multiplier** of the periodic orbit. σ is **unique**.

Spectral convergence



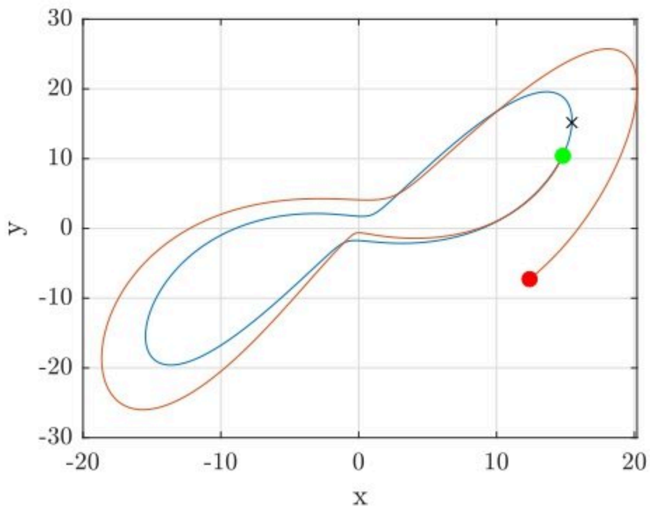


Zoom on the Floquet exponents (left) and the error of $\mu = 0.0465$ (right). Correct exponents: (-13.713, 0, 0.0465)

np	μ_{FD}	nt	$\mu_{Fourier}$	cnt	$\mu_{Chebyshev}$
		5	4.64931094e-02	10	6.199986e-02
16	4.4389e-02	7	4.64927338e-02	15	4.864546e-02
21	4.5295e-02	10	4.64927338e-02	20	4.648723e-02
31	4.5956e-02			30	4.649273e-02

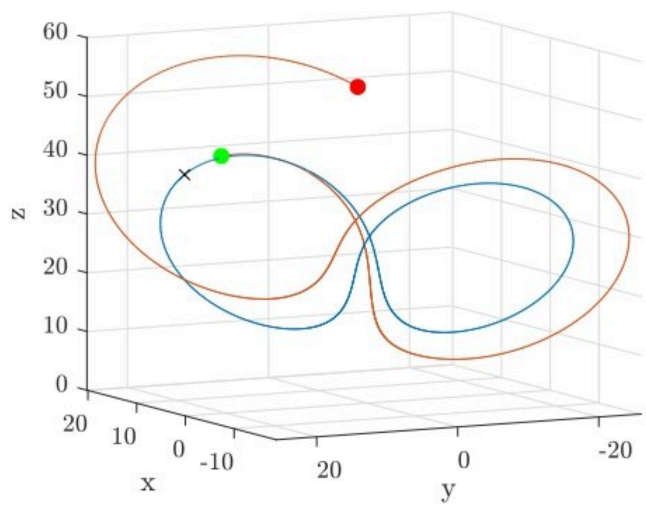
Most unstable Floquet exponent.

More complex orbit - $\rho = 28$, $\sigma = 10$, $\beta = 8/3$ | $\omega = 4.03$ $T = 1.56$



nt	μ_3	μ_2	μ_1
12	-1.4661e+01	6.8400e-15	9.9542e-01
16	-1.4661e+01	4.5433e-15	9.9466e-01
20	-1.4661e+01	-1.5859e-14	9.9465e-01

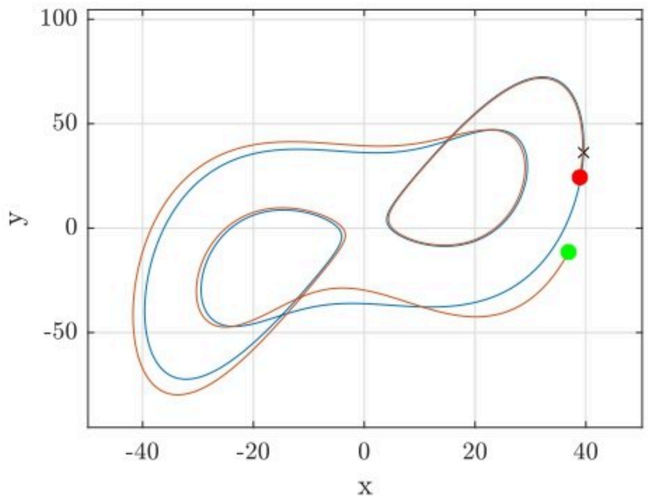
Hill's method



cnt	μ_3	μ_2	μ_1
30	-5.8174e+00	2.7928e-02	9.8702e-01
50	-9.4778e+00	2.4579e-03	9.9497e-01
70	-1.2319e+01	5.6688e-05	9.9466e-01
90	-1.4419e+01	1.1593e-06	9.9465e-01
110	-1.4694e+01	-1.5825e-07	9.9465e-01

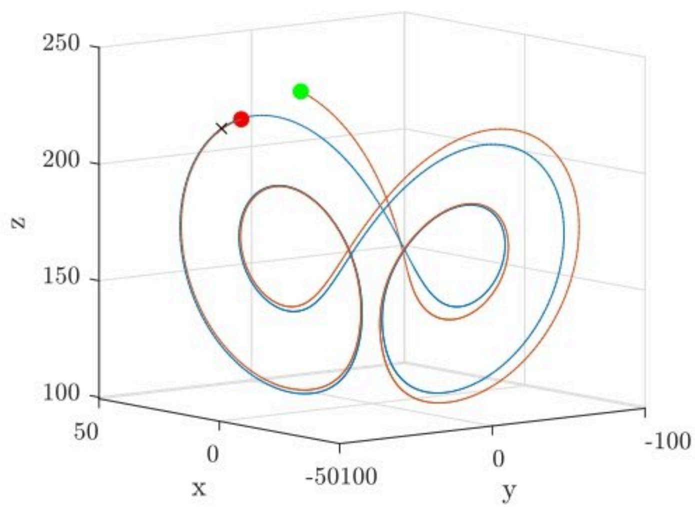
Chebyshev expression in time

Stable Lorenz butterfly orbit - $\rho = 160$, $\sigma = 10$, $\beta = 8/3 \mid \omega = 5.45$



nt	μ_3	μ_2	μ_1
20	-1.2469e+01	-1.1952e+00	-7.7734e-14
22	-1.2461e+01	-1.2055e+00	2.6118e-15
24	-1.2460e+01	-1.2060e+00	1.8593e-14
26	-1.2460e+01	-1.2067e+00	4.8119e-14
28	-1.2460e+01	-1.2068e+00	1.6459e-15

Hill's method



cnt	μ_3	μ_2	μ_1
50	-4.1469e+00	-2.5882e+00	-5.3971e-02
70	-8.9477e+00	-1.1366e+00	2.0596e-03
90	-1.1745e+01	-1.1925e+00	9.4145e-04
110	-1.4232e+01	-1.2049e+00	1.4470e-04
130	-1.2577e+01	-1.2067e+00	1.8836e-05
150	-1.2474e+01	-1.2069e+00	2.2549e-06

Chebyshev expression in time

What about complex Floquet multiplier? - Langford system

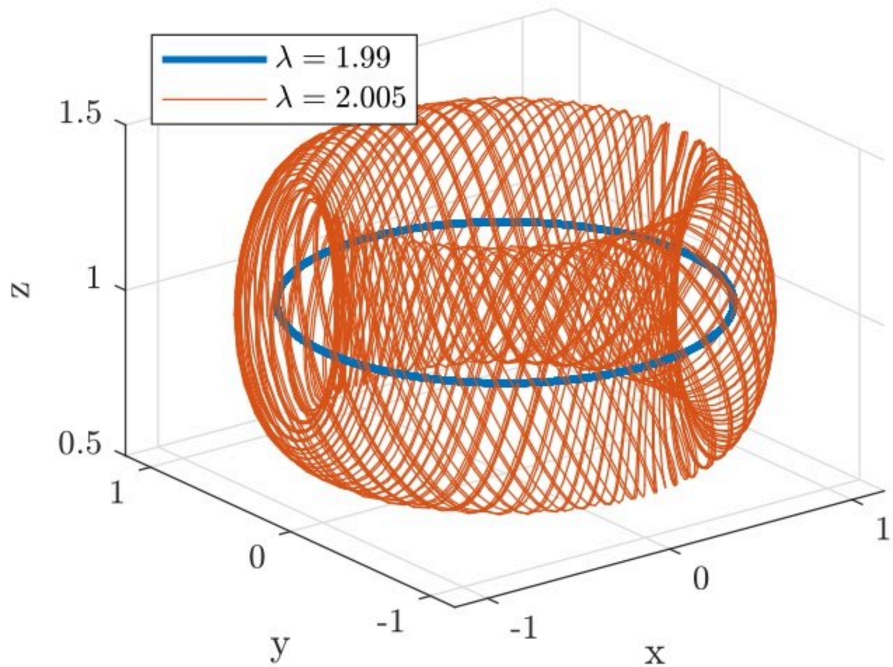
$$\dot{x} = (\lambda - 3)x - 0.25y + x\left(z + 0.2\left(1 - z^2\right)\right)$$

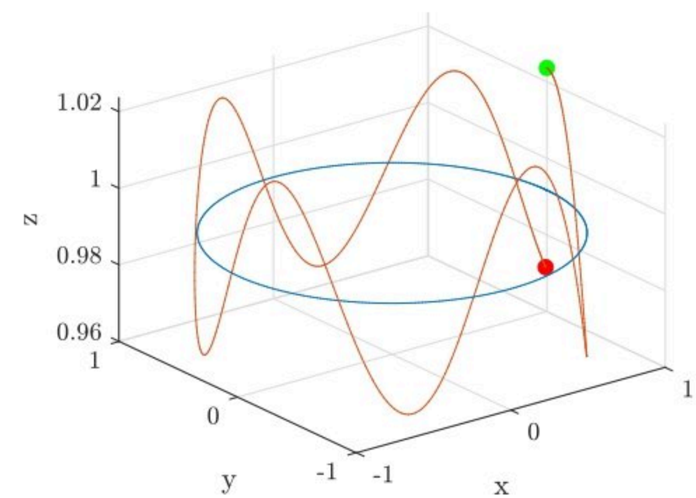
$$\dot{y} = (\lambda - 3)y + 0.25x + y\left(z + 0.2\left(1 - z^2\right)\right)$$

$$\dot{z} = \lambda z - \left(x^2 + y^2 + z^2\right)$$

Floquet exponents, circular orbit:

cnt	$\mu_{2,3}$ ($\lambda = 2.005$)
18	$2.7143 \times 10^{-2} \pm 1.0810i$
22	$1.1387 \times 10^{-2} \pm 1.1014i$
26	$1.0815 \times 10^{-2} \pm 1.1011i$
30	$1.0810355 \times 10^{-2} \pm 1.1011046i$
exact	$1.0810312 \times 10^{-2} \pm 1.1011045i$





Orbit plus an unstable eigenvector

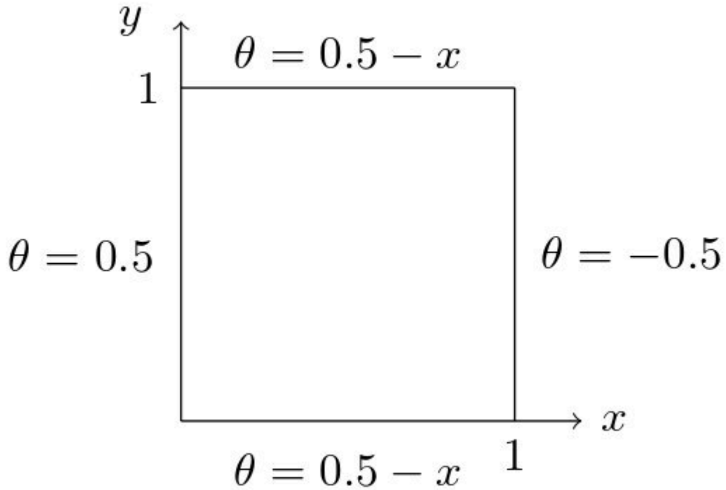
Differentially heated cavity

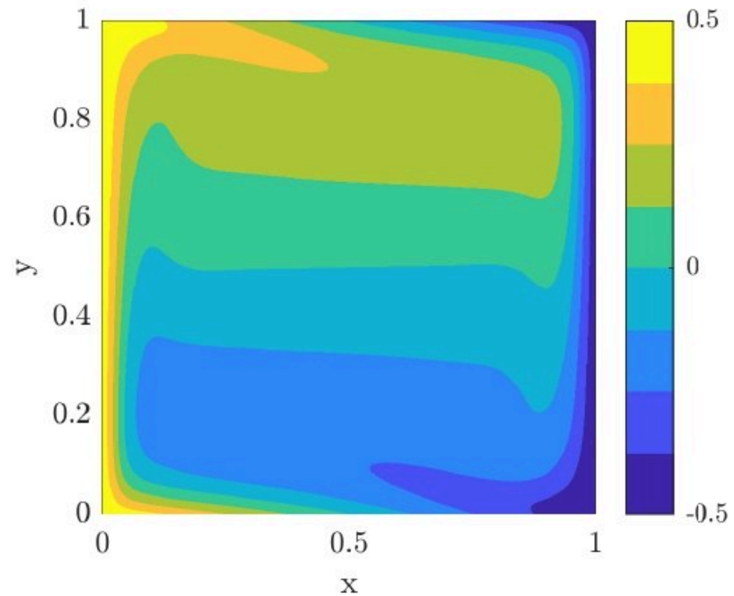
Navier-Stokes equations in Boussinesq approximation:

$$\frac{\partial \boldsymbol{u}}{\partial t}+(\boldsymbol{u}\cdot\nabla)\boldsymbol{u} = -\boldsymbol{\nabla} p+\frac{Pr}{\sqrt{Ra}}\boldsymbol{\nabla}^2\boldsymbol{u}+Pr\theta\boldsymbol{e}_y$$

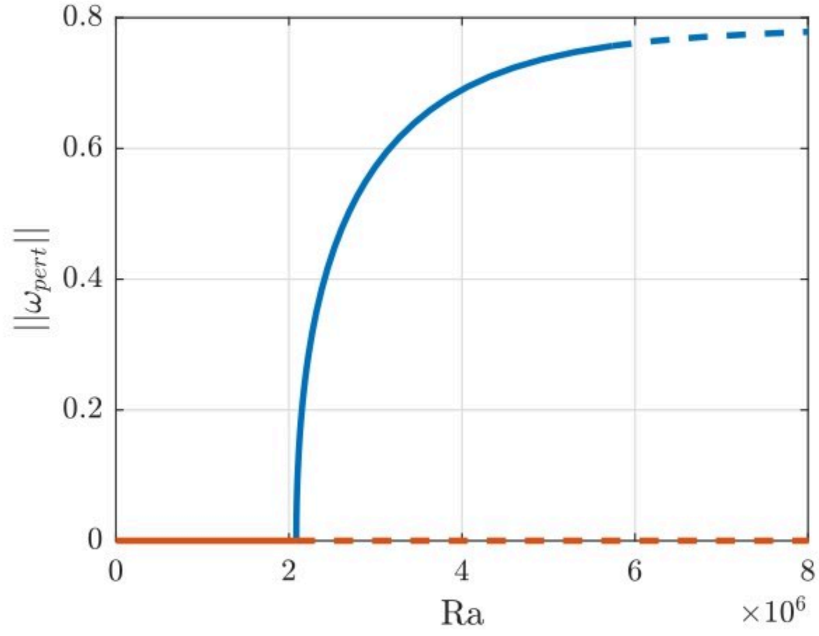
$$\frac{\partial \theta}{\partial t}+(\boldsymbol{u}\cdot\nabla)\theta = \frac{1}{\sqrt{Ra}}\boldsymbol{\nabla}^2\theta$$

$$\boldsymbol{\nabla}\cdot\boldsymbol{u}=0, \quad Pr=0.71$$





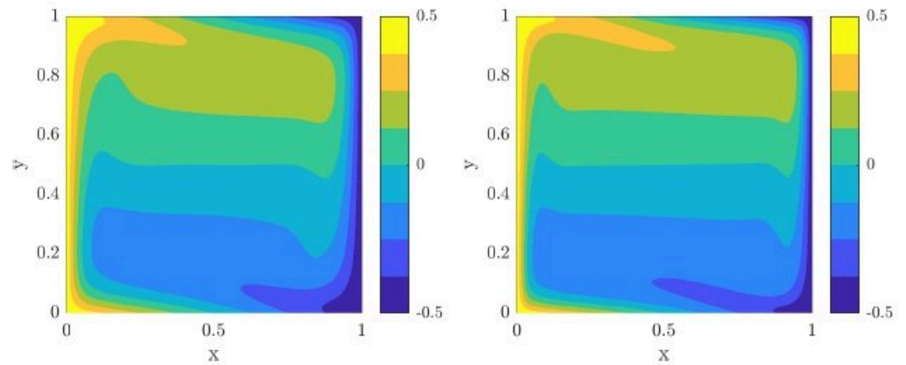
Temperature field of the base flow for $Ra = 2.2 \times 10^6$, mesh: nonuniform 128-128 cos-cos



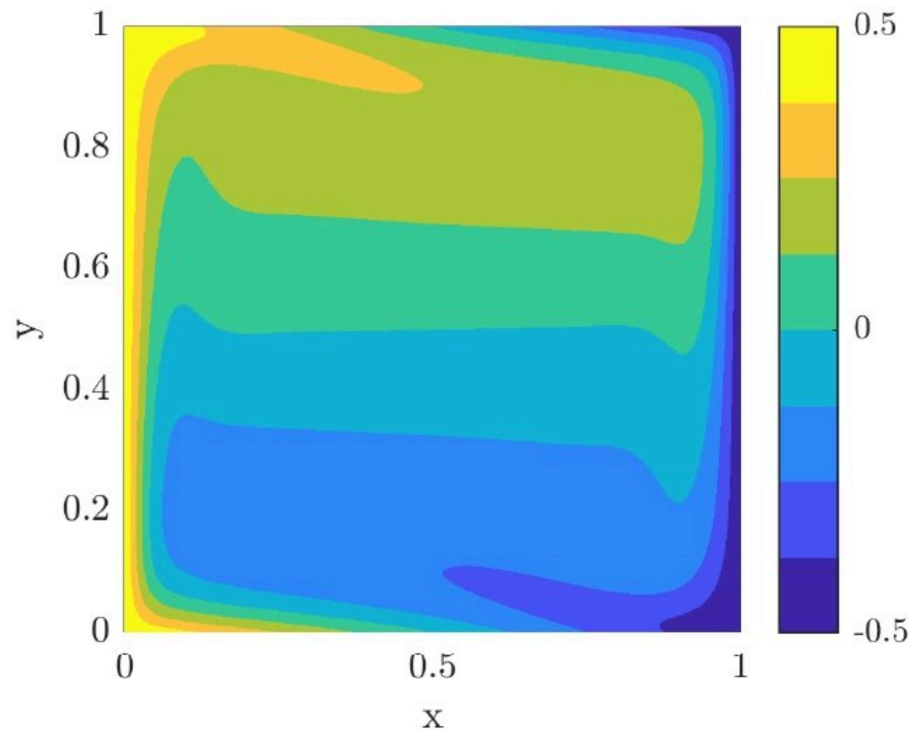
Bifurcation diagram

$$||\omega_{pert}|| = \sqrt{\iint |\omega - \omega_b|^2 dxdy}$$

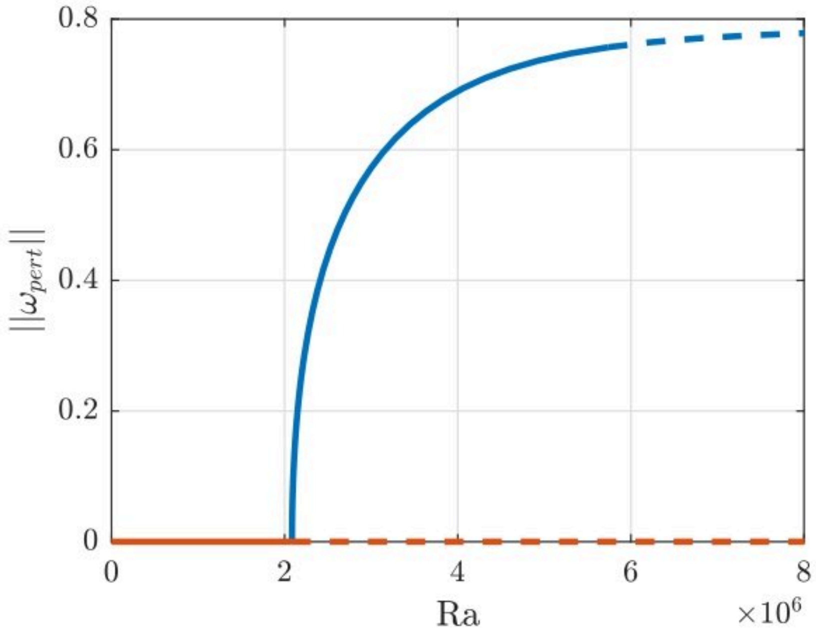
$$\omega = \frac{\partial u_y}{\partial x} - \frac{\partial u_x}{\partial y}$$



Base flow $Ra = 10^6$ and 5×10^6



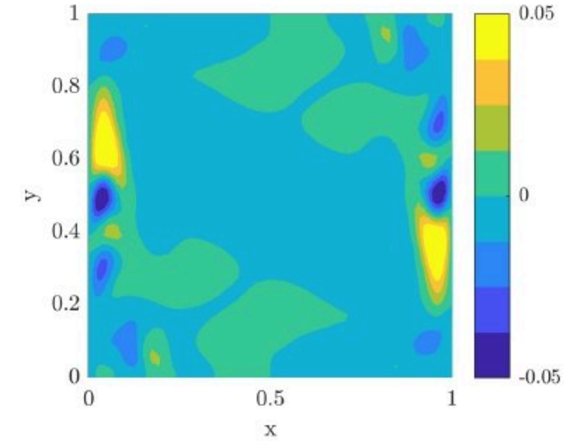
Saturated perturbation
$Ra = 3 \times 10^6$, angular freq. 1.38



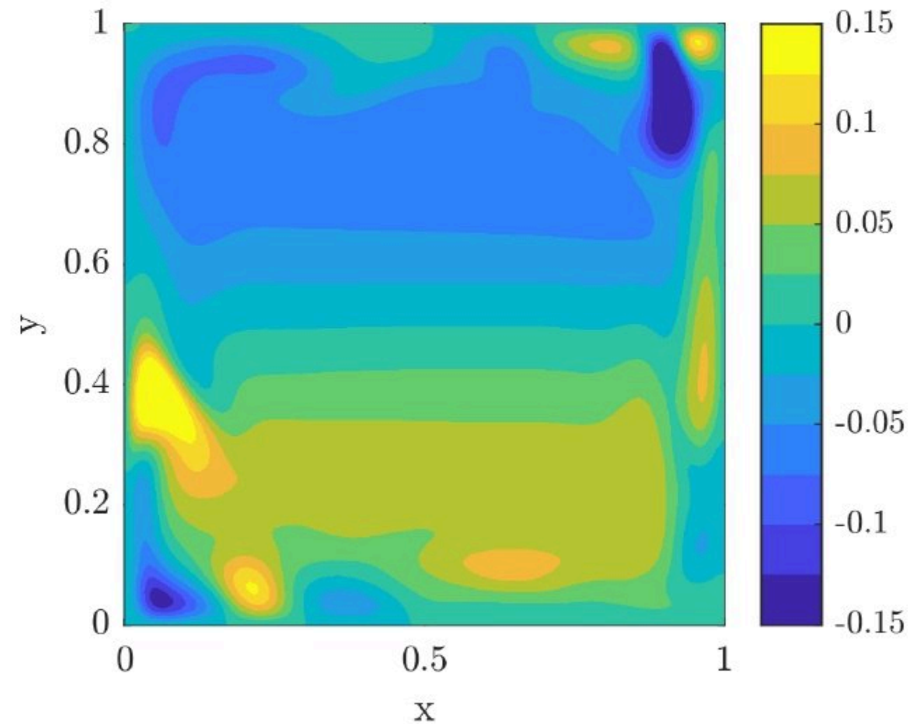
Bifurcation diagram

Ra	μ	$\lvert e^{\mu T} \rvert$
5.0×10^6	-3.17e-03+-2.33e-01i	9.855e-01
5.5×10^6	-1.39e-03+-2.32e-01i	9.936e-01
6.0×10^6	-1.84e-04+-2.32e-01i	9.991e-01
6.5×10^6	5.64e-04+2.32e-01i	1.003e+00
7.0×10^6	9.52e-04+2.32e-01i	1.004e+00

Stability with 26 Chebyshev modes



Orbit's eigenvector $Ra = 5 \times 10^6$



Time evolution $Ra = 7 \times 10^6$

Summary

Chebyshev polynomials:

- can be successfully employed in description of periodic dynamics
- offer lower accuracy than the Fourier basis but
- the stability of the orbit can be unambiguously interpreted

Best practice:

- use Fourier basis to track the orbit and Chebyshev basis to identify the physically relevant Floquet exponents of the Hill's matrix